Wi-Fi Module



Category: **Electrical & Electronic Components & Kits** Type: **Short SOP** Priority: **Medium**

1 labs

Covers these lab items: Wi-Fi Module (ESP8266/ESP32)

What it is used for

A Wi-Fi module such as ESP8266 or ESP32 enables compatible electronics projects to communicate wirelessly. Configure the module using its documentation and verify power, wiring, and network settings before testing.

How to use

1. Identify the module version and check its voltage, pinout, and interface requirements.
2. Inspect the module, development board, USB cable, and connecting wires for damage.
3. Connect the module to a compatible power supply and controller using the documented wiring.
4. Install the required development environment, drivers, and approved firmware tools.
5. Configure the wireless network credentials and application settings securely.
6. Upload or run a simple test program and verify connection status and data exchange.
7. Disconnect power before changing wiring and record the working configuration.

⚠ Safety

- Use the specified supply voltage and avoid short circuits between pins.
- Do not connect GPIO pins directly to incompatible voltage levels.
- Never expose the module to liquids or conductive debris.
- Protect network credentials and avoid sharing sensitive keys in public code.

Care & storage

- Handle the module by its edges and store it in an antistatic, dry environment.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.